

Barrel / Cylinder Detection in 3D Point Clouds

A Survey of Methods and their Suitability for Occluded Detection

Master's-thesis literature backbone · Bharath · 2026-05-30

Living document — regenerated as the method list and references evolve.

1 Purpose and scope

This document is the literature backbone for the thesis comparison of barrel/cylinder detection methods in 3D point clouds. The aim is **empirical and comparative**: to determine *which methods detect barrels most accurately* — particularly under occlusion and partial burial — and to *characterise the strengths, weaknesses, and operating regimes* of each method family.

It is deliberately **not** committed to proving a single predetermined hypothesis. Any trade-offs we observe (for example, how precision and recall move as occlusion increases, or how camera+LiDAR fusion compares with LiDAR-only methods) are *findings to be measured*, not assumptions to be defended.

The comparison is made fair by a shared data / I/O contract: every method reads the same clouds and emits the same prediction schema, and a single evaluator scores them (precision / recall / F1, radius and centre error, axis-angle error, runtime, and annotation / training cost kept as a column so geometric and learned methods compare fairly). Ground-truthed data — including the *true per-barrel occlusion percentage* — will come from NVIDIA Isaac Sim, with validation on a small set of real Asus Xtion scans. The current baseline is the 3DTK randomized-Hough `detectCylinder`.

Reading the survey. Methods already named in the project plan are marked with a dagger (†); the rest were surfaced by a literature search on 2026-05-30. Every method maps to one or more entries in the bibliography (§6) via the short key in the *Key refs* column.

2 How to read the occlusion-suitability rating

Each method carries a 1–5 rating for **suitability to occluded / partial-view barrel detection** — the central difficulty of this thesis. The rating is an informed prior from the literature whose only job is to *prioritise experiments*; the thesis will replace it with measured numbers. It weighs five criteria:

1. **Partial-evidence detection** — can it fire from a partial arc or sparse returns?
2. **Robustness to fragmentation** — does missing surface break its core mechanism (region growth, clustering, connectivity)?
3. **Learnability of occlusion** — can occluded cases be learned from data, and is such data affordable here?
4. **Second-modality / completion support** — does a camera or a completion stage recover the object when LiDAR is occluded?
5. **Localisation under clutter** — can it separate adjacent / overlapping barrels?

Scale. 5 Very High purpose-built for occlusion | 4 High handles it well (often data-dependent) | 3 Moderate partial views OK, degrades with heavy occlusion/clutter | 2 Low-Mod fragile under occlusion | 1 Low unsuited.

3 Methods comparison table

#	Method	Core idea	Occlusion fit	Why (behaviour under occlusion / clutter)	Key refs
A. LiDAR-only — classical / geometric					
1	Euclidean clustering + per-cluster fit †	Remove ground/planes, cluster the rest (DBSCAN/Euclidean), fit one cylinder per cluster.	★★☆☆☆ Low-Mod	Occlusion and adjacency split or merge clusters; touching barrels collapse into one. Fast and simple, but the proposer is the weak link. (<i>Project's current proposer.</i>)	project pipeline
2	Region growing + normal curvature †	Grow surface regions under a smoothness / curvature constraint, then fit.	★★☆☆☆ Low-Mod	Needs a contiguous smooth surface; occlusion fragments the shell into disconnected patches, giving broken / under-segmented regions.	Rabbani'06, Moradi'21
3	Randomized 2-step Hough — 3DTK detectCylinder (<i>baseline</i>) †	2D Hough on the Gaussian sphere of normals for the axis direction, then 3D Hough for centre + radius.	★★★★☆ Moderate	A partial arc still votes and axis voting tolerates partial views; but heavy occlusion thins the accumulator and clutter creates spurious peaks. Lock radius to the drum band; tune accumulator resolution + sampling.	Rabbani'05, Borrmann'11
4	RANSAC cylinder fit (PCL) †	Sample minimal point+normal sets, hypothesise a cylinder, score inliers; PCA for axis.	★★★★☆ Moderate	Tolerates outliers and fits from a partial arc, but inlier thresholds are noise-sensitive and clutter spawns competing hypotheses; needs a proposer to localise.	PCL; Nurnnabi'19
5	Efficient RANSAC — multi-primitive	Accelerated RANSAC detecting planes / spheres / cylinders / cones / tori at once.	★★★★☆ Moderate	Robust to heavy noise/outliers and scales well; partial cylinders detectable, but over-detects primitives, so it needs post-filtering to the barrel prior.	Schnabel'07
6	Robust least-squares cylinder fit (<i>fit stage</i>)	Faithful / robust (M-estimator, robust PCA) least-squares fit of cylinder parameters to inliers.	★★★★☆ Moderate	Degrades gracefully on partial arcs and resists outliers — but it is a <i>fit</i> , not a detector; quality is bounded by the proposer that hands it points.	Lukács'98, Nurnnabi'19
7	Connectivity / projection-based detection	Project points along hemisphere directions; detect circular projections via connected components. Little tuning, arbitrary orientation.	★★★★☆ Moderate	Noise-robust and parameter-light; circular-projection evidence survives partial views, but connected-component reasoning can break under fragmentation.	Araújo'20
B. LiDAR-only — learned					
8	PointNet++ (segment → fit) †	Hierarchical point-set network segments barrel points; metric fit recovers pose/radius.	★★★★☆ High	Learns partial-shape cues, so it can flag occluded barrels <i>if trained on occluded examples</i> — exactly BarrelNet's recipe. Cost: annotation + sim-to-real gap.	Qi'17; cf. Yan'24
9	VoteNet — deep Hough voting	Seed points vote for object centres; cluster votes into proposals.	★★★★☆ High	Centre-voting is built for partial / occluded objects — visible surface points still vote toward the hidden centroid. Needs 3D-box training labels.	Qi'19
10	Voxel / pillar detectors — PointPillars, SECOND, PV-RCNN †	Voxel/pillar encoder + sparse-conv backbone + detection head; PV-RCNN adds point-voxel refinement.	★★★★☆ High	Trained end-to-end on occlusion-rich benchmarks; learn to detect from partial evidence. Strong accuracy; cost = annotated data + compute.	Lang'19, Shi'20
11	CenterPoint — center-based detection †	Detect object centres as BEV heatmap peaks; regress box / size / orientation.	★★★★☆ High	The centre representation is robust to partial observation and crowding; SOTA-class on driving data. Same data / compute caveat.	Yin'21
12	BtcDet — “Behind the Curtain” (occlusion-aware)	Learn shape priors, estimate occupancy in occluded / missed regions, complete shapes, then detect.	★★★★★ Very High	Purpose-built for occlusion — explicitly reasons about what is hidden. Highest relevance to the occlusion question; higher engineering complexity.	Xu'22

#	Method	Core idea	Occlusion fit	Why (behaviour under occlusion / clutter)	Key refs
13	Learned primitive fitting — SPFN / Point2Cyl / PrimiTect	Predict per-point primitive membership/type/normals (SPFN), extrusion-cylinder decomposition (Point2Cyl), or continuous Hough voting (PrimiTect).	★★★★☆ Moderate	Beat RANSAC on clean / CAD data and give principled cylinder params; partial-view + sim-to-real generalisation for field barrels is unproven. Point2Cyl targets CAD extrusions.	Li'19, Uy'22, Sommer'20
14	Shape completion → fit — PCN (<i>pre-stage</i>)	Complete the partial cloud (encoder-decoder), then fit / detect on the completed shell.	★★★★☆ High	Directly attacks occlusion by hallucinating the missing shell before fitting; adds a learned stage with its own failure modes and training needs.	Yuan'18
C. Camera + LiDAR fusion					
15	Camera-first frustum — YOLO → frustum → fit; Frustum-PointNets / -ConvNet †	2D image detector proposes a region; back-project to a 3D frustum; segment + fit inside it.	★★★★☆ High	The camera still sees a barrel when its LiDAR return is sparse / occluded ⇒ strong recall under occlusion. Failure modes: object must be visible in-image, and extrinsic-calibration error degrades the frustum.	Qi'18, Wang'19
16	LiDAR-first → image verification †	Generate 3D proposals from LiDAR, project to the image, verify / score with the 2D detector (decision-level fusion).	★★★★☆ High	Image confirmation suppresses false positives and rescues low-confidence detections; but the proposal still needs enough LiDAR points on the occluded object.	Pang'20 (CLOCs)
17	Point decoration — PointPainting / PointAugmenting †	Paint each LiDAR point with image semantics / CNN features, then run any LiDAR detector.	★★★★☆ High	Semantic painting separates barrels from clutter and helps thin returns; effectiveness still bounded by having points on the object and by calibration.	Vora'20
18	Deep multimodal fusion — TransFusion / BEVFusion / EPNet / MVX-Net	Feature-level fusion of camera + LiDAR (transformer queries / unified BEV).	★★★★★ Very High	SOTA on occluded driving benchmarks and comparatively calibration-robust (TransFusion's soft association); cost = training data, compute, integration complexity.	Bai'22, Liu'23

† = method already named in the project plan (CLAUDE.md / memory). (*baseline*) = current working baseline. Rating scale defined in §2.

4 Closest prior work: occlusion and barrels specifically

A targeted search surfaced work sitting almost exactly on this thesis:

- **BarrelNet** (Yan et al., OCEANS 2024) is the nearest neighbour: a PointNet-style network trained on *synthetically generated* cylinder point clouds that explicitly model camera-viewpoint occlusion *and* burial in the seafloor. It regresses 6-DOF pose, radius, and a *burial fraction*, and *beats a classical least-squares cylinder fit* on synthetic tests, with qualitative ROV-footage generalisation at a Southern-California DDT-barrel dump site. This validates the core thesis bet — synthetic occluded / buried training plus comparison against a geometric baseline — and is the single most important paper to position against.
- **Behind the Curtain / BtcDet** (Xu et al., AAAI 2022) makes occlusion a first-class citizen for generic LiDAR detection: it learns shape priors and predicts occupancy in occluded regions *before* detecting. It is the strongest template for an occlusion-aware learned detector here.
- **Shape completion** (PCN, Yuan et al., 3DV 2018) is an orthogonal lever: complete the occluded shell, *then* fit — a drop-in pre-stage for any geometric method.
- **Occlusion analysis** — OccAM’s Laser (Schinagl et al., CVPR 2022) produces occlusion-based attribution maps for 3D detectors, useful for *explaining* why a method’s recall falls as occlusion rises.
- **Buried-object context beyond LiDAR** — integrated geophysical surveys (magnetometry / electromagnetic induction) detect buried steel drums where optical / LiDAR sensing cannot. Out of scope as a detection method, but relevant when framing the buried-barrel problem and its sensing limits.

Takeaway. Occlusion-aware learned detection and synthetic occluded training are established, and in the one published barrel-specific study a learned method beats classical fitting. The thesis should therefore treat a learned occlusion-aware method (a BarrelNet-style PointNet++, or BtcDet) as a *serious contender* against the 3DTK Hough baseline, not an afterthought — while the geometric methods remain the zero-training reference point.

5 Suggested shortlist for the comparison

Balancing relevance, occlusion-suitability, and implementation cost, a defensible first comparison set is:

- **Geometric baselines:** 3DTK randomized Hough (*already running*) and a RANSAC / efficient-RANSAC cylinder fit — cheap, no training, the reference point.
- **Learned LiDAR:** a PointNet++ / BarrelNet-style detector trained on Isaac-Sim occluded / buried barrels; optionally BtcDet as the occlusion specialist.
- **Fusion:** one camera-first frustum pipeline (YOLO → frustum → fit) to probe recall under occlusion, with a calibration-perturbation analysis as the fusion stress test.

This spans all three families, includes the occlusion specialist, and keeps the geometric reference — enough to answer *which method works best, and why, per occlusion regime*.

6 References (grouped, with links)

A. Classical / geometric cylinder detection & fitting

- **[Rabbani’05]** T. Rabbani, F. van den Heuvel. *Efficient Hough Transform for Automatic Detection of Cylinders in Point Clouds*. ISPRS Workshop “Laser Scanning 2005”. (2-step Gaussian-sphere + 3D Hough — the method 3DTK `detectCylinder` implements.) [\[pdf\]](#)
- **[Borrmann’11]** D. Borrmann, J. Elseberg, K. Lingemann, A. Nüchter. *The 3D Hough Transform for Plane Detection in Point Clouds: A Review and a New Accumulator Design*. 3D Research 2(2), 2011. (3DTK Hough lineage.) [\[pdf\]](#)
- **[Schnabel’07]** R. Schnabel, R. Wahl, R. Klein. *Efficient RANSAC for Point-Cloud Shape Detection*. Computer Graphics Forum 26(2):214–226, 2007. [\[doi\]](#)
- **[Lukács’98]** G. Lukács, R. Martin, D. Marshall. *Faithful Least-Squares Fitting of Spheres, Cylinders, Cones and Tori for Reliable Segmentation*. ECCV 1998, LNCS 1406. [\[link\]](#)
- **[Rabbani’06]** T. Rabbani, F. van den Heuvel, G. Vosselman. *Segmentation of Point Clouds Using Smoothness Constraint*. ISPRS Commission V Symposium, 2006. (Region growing by normal smoothness.)

- **[Nurunnabi’19]** A. Nurunnabi, Y. Sadahiro, R. Lindenbergh, D. Belton. *Robust Cylinder Fitting in Laser Scanning Point Cloud Data*. Measurement 138:632–651, 2019. [\[link\]](#)
- **[Araújo’20]** A. M. C. Araújo, M. M. Oliveira. *Connectivity-based Cylinder Detection in Unorganized Point Clouds*. Pattern Recognition 100:107161, 2020. [\[link\]](#) · [\[code\]](#)
- **[Moradi’21]** S. Moradi, D. Laurendeau, C. Gosselin. *Multiple Cylinder Extraction from Organized Point Clouds*. Sensors 21(22):7630, 2021. [\[link\]](#)

B. Learned LiDAR-only detectors

- **[Qi’17]** C. R. Qi, L. Yi, H. Su, L. J. Guibas. *PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space*. NeurIPS 2017. [\[arXiv:1706.02413\]](#)
- **[Qi’19]** C. R. Qi, O. Litany, K. He, L. J. Guibas. *Deep Hough Voting for 3D Object Detection in Point Clouds (VoteNet)*. ICCV 2019. [\[arXiv:1904.09664\]](#)
- **[Lang’19]** A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, O. Beijbom. *PointPillars: Fast Encoders for Object Detection from Point Clouds*. CVPR 2019. [\[arXiv:1812.05784\]](#)
- **[Shi’20]** S. Shi, C. Guo, L. Jiang, et al. *PV-RCNN: Point-Voxel Feature Set Abstraction for 3D Object Detection*. CVPR 2020. [\[arXiv:1912.13192\]](#) (see also SECOND: Yan et al., Sensors 2018, and PointRCNN: Shi et al., CVPR 2019.)
- **[Yin’21]** T. Yin, X. Zhou, P. Krähenbühl. *Center-based 3D Object Detection and Tracking (CenterPoint)*. CVPR 2021. [\[arXiv:2006.11275\]](#)
- **[Xu’22]** Q. Xu, Y. Zhong, U. Neumann. *Behind the Curtain: Learning Occluded Shapes for 3D Object Detection (BtcDet)*. AAAI 2022. [\[arXiv:2112.02205\]](#)
- **[OpenPCDet]** OpenPCDet — open-source toolbox implementing SECOND, PointPillars, PV-RCNN, CenterPoint, etc. [\[github\]](#)

C. Learned primitive fitting & shape completion

- **[Li’19]** L. Li, M. Sung, A. Dubrovina, L. Yi, L. J. Guibas. *Supervised Fitting of Geometric Primitives to 3D Point Clouds (SPFN)*. CVPR 2019. [\[arXiv:1811.08988\]](#)
- **[Uy’22]** M. A. Uy, Y.-Y. Chang, M. Sung, P. Goel, J. Lambourne, T. Birdal, L. J. Guibas. *Point2Cyl: Reverse Engineering 3D Objects from Point Clouds to Extrusion Cylinders*. CVPR 2022. [\[arXiv:2112.09329\]](#)
- **[Sommer’20]** C. Sommer, Y. Sun, E. Bylow, D. Cremers. *PrimiTect: Fast Continuous Hough Voting for Primitive Detection*. ICRA 2020. [\[arXiv:2005.07457\]](#)
- **[Yuan’18]** W. Yuan, T. Khot, D. Held, C. Mertz, M. Hebert. *PCN: Point Completion Network*. 3DV 2018. [\[arXiv:1808.00671\]](#)

D. Camera–LiDAR fusion

- **[Qi’18]** C. R. Qi, W. Liu, C. Wu, H. Su, L. J. Guibas. *Frustum PointNets for 3D Object Detection from RGB-D Data*. CVPR 2018. [\[arXiv:1711.08488\]](#)
- **[Wang’19]** Z. Wang, K. Jia. *Frustum ConvNet: Sliding Frustums to Aggregate Local Point-Wise Features for Amodal 3D Object Detection*. IROS 2019. [\[arXiv:1903.01864\]](#)
- **[Vora’20]** S. Vora, A. H. Lang, B. Helou, O. Beijbom. *PointPainting: Sequential Fusion for 3D Object Detection*. CVPR 2020. [\[arXiv:1911.10150\]](#) (see also PointAugmenting, Wang et al., CVPR 2021.)
- **[Sindagi’19]** V. A. Sindagi, Y. Zhou, O. Tuzel. *MVX-Net: Multimodal VoxelNet for 3D Object Detection*. ICRA 2019. [\[arXiv:1904.01649\]](#)
- **[Huang’20]** T. Huang, Z. Liu, X. Chen, X. Bai. *EPNet: Enhancing Point Features with Image Semantics for 3D Object Detection*. ECCV 2020. [\[arXiv:2007.08856\]](#)
- **[Pang’20]** S. Pang, D. Morris, H. Radha. *CLOCs: Camera-LiDAR Object Candidates Fusion for 3D Object Detection*. IROS 2020. [\[arXiv:2009.00784\]](#)
- **[Bai’22]** X. Bai, Z. Hu, X. Zhu, et al. *TransFusion: Robust LiDAR-Camera Fusion for 3D Object Detection with Transformers*. CVPR 2022. [\[arXiv:2203.11496\]](#)
- **[Liu’23]** Z. Liu, H. Tang, A. Amini, et al. *BEVFusion: Multi-Task Multi-Sensor Fusion with Unified Bird’s-Eye-View Representation*. ICRA 2023. [\[arXiv:2205.13542\]](#)

E. Occlusion-robust & barrel / buried-object specific

- [Yan’24] J. Yan, C. Talegaonkar, N. Antipa, E. Terrill, S. Merrifield. *Pose Estimation of Buried Deep-Sea Objects using 3D Vision Deep Learning Models* (BarrelNet). OCEANS 2024 — Halifax. [\[arXiv:2410.01061\]](#) — **closest prior work**.
- [Xu’22] BtcDet — “Behind the Curtain” (occlusion-aware detection); listed under §B above. [\[arXiv:2112.02205\]](#)
- [Schinagl’22] D. Schinagl, G. Krispel, H. Possegger, P. M. Roth, H. Bischof. *OccAM’s Laser: Occlusion-based Attribution Maps for 3D Object Detectors on LiDAR Data*. CVPR 2022. [\[arXiv:2204.06577\]](#)
- [Geophys’10] *Integrated Geophysical Measurements on a Test Site for Detection of Buried Steel Drums*. 2010. (Non-LiDAR buried-drum context: magnetometry / EMI.) [\[arXiv:1009.5555\]](#)

Citation note. References were compiled via literature search on 2026-05-30. Authors, venues, and arXiv identifiers were checked against the linked sources, but please re-verify each entry against its link before quoting it in the final thesis. This file lives in `researchwrite/` and is regenerated as the survey evolves; the companion index is the `researchwrite-materials` memory.

7 Off-the-shelf implementations (GitHub) — what to reuse

The plan is to **reuse code, not reimplement**. For each method (keyed to the table in §3) the table below gives the *recommended* repository — the most-starred, actively maintained option — with lighter or alternative repos in *Notes*. Star counts (★), language and licence are as of 2026-05-30.

Two caveats. (i) **BarrelNet** [Yan’24], the closest prior work, has *no public code release* — it must be reproduced on top of a PointNet++ repo. (ii) A few official repos are *archived* (read-only on GitHub) but still compile and run.

Toolchain shortcut. Standardise on [open-mmlab/OpenPCDet](#). PointPillars, SECOND, PV-RCNN and CenterPoint all ship inside it; BtcDet is built on it; and CLOCs and the recommended PointPainting repo plug into it. Adopting one framework puts almost all the learned-LiDAR methods plus several fusion methods in a *single* codebase — little to no from-scratch work, one training/eval harness, and an easy bridge to the project’s `predictions.json` contract.

Method (cf. §3)	Recommended GitHub repo	★ Stars	Lang / Licence	Notes (alternatives & caveats)
A. LiDAR-only — classical / geometric				
Euclidean clustering + fit	PointCloudLibrary/pcl	11.0k	C++ / BSD	<code>EuclideanClusterExtraction</code> + <code>SACSegmentation</code> (cylinder model) in one library. Python route: isl-org/Open3D (13.6k, MIT) for clustering, paired with a cylinder fit.
Region growing + curvature	PointCloudLibrary/pcl	11.0k	C++ / BSD	<code>pcl::RegionGrowing</code> (smoothness / curvature constraint) — the [Rabani'06] method.
Randomized 2-step Hough — 3DTK (<i>baseline</i>)	JMUWRobotics/3DTK	131	C++	Already vendored. Maintained mirror; provides <code>detectCylinder</code> .
RANSAC cylinder fit	leomariga/pyRANSAC-3D	653	Python / Apache	Pure-Python <code>Cylinder()</code> — easiest pull for this stack. C++ reference: PCL <code>SACMODEL_CYLINDER</code> (11.0k).
Efficient RANSAC (multi-primitive)	CGAL/cgal	5.9k	C++ / GPL+comm.	“Point-Set Shape Detection” = the [Schnabel'07] algorithm. Lighter standalone: LiangliangNan/Easy3D (1.6k, GPL).
Robust least-squares cylinder fit	xingjiepan/cylinder_fitting	53	Python / BSD	pip-installable least-squares cylinder fit (Eberly). <i>Fit stage only</i> — needs a proposer.
Connectivity / projection detection	abner-math/CylinderDetection	39	C++ / GPL	Author's official code for [Araújo'20].
B. LiDAR-only — learned				
PointNet++ (basis for BarrelNet)	yanx27/Pointnet_Pointnet2_pytorch	4.9k	Python / MIT	Most-popular PyTorch PointNet++. BarrelNet [Yan'24] has no public code — reproduce here. Alt: erikwijmans/Pointnet2_PyTorch (1.8k).
VoteNet	facebookresearch/votenet	1.8k	Python / MIT	Official deep-Hough-voting detector. Repo <i>archived</i> (read-only) but runs.
PointPillars / SECOND / PV-RCNN	open-mmlab/OpenPCDet	5.6k	Python / Apache	All three (+ Part-A2, etc.) in one maintained hub. Alt: open-mmlab/mmdetection3d (6.4k).
CenterPoint	tianweiy/CenterPoint	2.2k	Python / MIT	Official center-based detector; also bundled inside OpenPCDet.
BtcDet — occlusion-aware	Xharlie/BtcDet	201	Python / Apache	Official; built on OpenPCDet 0.3.0 — drops into the same toolchain. Best occlusion specialist.
Learned primitive fitting	lingxiaoli94/SPFN	184	Python / MIT	SPFN (multi-primitive). Also mikacuy/point2cyl (65, extrusion cylinders) and c-sommer/primitect (33, C++).
Shape completion (PCN, pre-stage)	wentaoyuan/pcn	482	Python / MIT	Official (TensorFlow). PyTorch reimpl: qinglew/PCN-PyTorch (212).
C. Camera + LiDAR fusion				
Camera-first frustum	charlesq34/frustum-pointnets	1.7k	Python / Apache	Official Frustum-PointNets (TF). F-ConvNet: Gorilla-Lab-SCUT/frustum-convnet (247, PyTorch).
LiDAR-first → image verification	pangsu0613/CLOCs	405	Python / MIT	Decision-level fusion on top of SECOND / PV-RCNN (OpenPCDet-compatible).
Point decoration (PointPainting)	Song-Jingyu/PointPainting	314	Python / MIT	mmsegmentation + OpenPCDet, maintained. Alt: AmrElsersy/PointPainting (170, real-time BiSeNetv2).
Deep multimodal fusion	mit-han-lab/bevfusion	3.2k	Python / Apache	Most-popular fusion repo; <i>archived</i> but runs. Alt: Xuyang-Bai/TransFusion (760); MVX-Net via mmdetection3d (6.4k); happinesslz/EPNet (254).

Star counts are a popularity snapshot (GitHub API, 2026-05-30) and drift over time; “archived” repos are read-only but still clone/build. Where a method has several implementations, the most-starred actively-maintained one is recommended and the rest are listed as alternatives.